



Review

Structural and dynamical properties of Fe₇₈Si₉B₁₃ alloy during rapid quenching by first principles molecular dynamic simulationJingyu Qin^{a,*}, Tingkun Gu^{a,b}, Lei Yang^a^aThe Key Laboratory of Liquid Structure and Heredity of Materials, Ministry of Education, Shandong University, Jinan 250061, China^bThe School of Electrical Engineering of Shandong University, Jinan 250061, China

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ABSTRACT

Structural evolution of the Fe₇₈Si₉B₁₃ alloy during rapid quenching was investigated by *ab initio* molecular dynamics simulation. The second peak splitting has been perceived even at 1473 K in the partial pair correlation functions though not in the total pair correlation function. The (0, 3, 6, 0) polyhedra are abundant in the liquid state while the distorted (0, 3, 6, 0) polyhedra are the featured local structure around B atoms in the amorphous state. The diffusion coefficients of the three elements are evaluated to understand the dynamics of quenching. From 1173 to 873 K the three coefficients are coherent. We think this temperature range corresponds to the supercooled liquid region, and 873 K serves as the glass transition temperature.

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1. Introduction

Fe-based amorphous alloys are important materials which have potential applications. Multi-component amorphous Fe-based steels are paid more attention recently [1,2]. In order to understand the mechanisms of both the formation and crystallization, the knowledge on the structure and its evolution with temperature is important. However, the researchers face intrinsic trouble to investigate the partial structures of multi-component alloys through experimental methods. The classical molecular dynamics simulation is not convenient in most cases to investigate the partial structures due to the lack of proper interatomic potentials. Fortunately, the *ab initio* molecular dynamics simulation (AIMD) provides an easy and efficient way to deal with the partial structural information of the multi-component alloys.

Fe–Si–B alloys are widely studied for many years [3], and are important basic alloys for the development of multi-component magnetic alloys. So it is suitable to employ one of such alloys as a model to study the structural evolution during rapid solidification.

Previously we found that in the Fe₇₈Si₉B₁₃ alloy the local structure around Si atoms may exhibit rather large difference between the liquid and amorphous structure in the Fe₇₈Si₉B₁₃ alloy by AIMD [4]. In this paper, we will present detailed information on the structural evolution of the Fe₇₈Si₉B₁₃ alloy with temperature. We can also monitor the change of the diffusion coefficients with temperature. In this way we tried to perceive the supercooled liquid region and glass transition. We hope the results from such a model alloy could strengthen the understanding of the multiple component alloys.

2. Methodology

The AIMD simulations were carried out using the most recent version of the Vienna *ab initio* simulation package (VASP) [5] in which the interactions between the ions and electrons are described by the projector augmented-wave method [6]. Simulations were performed using a gradient corrected energy functional [7]. The plane-wave basis set contained components with energies up to 319.0 eV which is the default value supplied by the pseudopotentials of the VASP package. We also tested a cutoff energy which is 20% larger than the default value at 1473 K; the resulted six independent partial pair correlation

* Corresponding author. Tel.: +86 531 88392810.

E-mail address: qin jy@sdu.edu.cn (J. Qin).

functions (PCFs), i.e. $g_{\text{FeFe}}(r)$, $g_{\text{FeSi}}(r)$, $g_{\text{FeB}}(r)$, $g_{\text{SiSi}}(r)$, $g_{\text{BSi}}(r)$ and $g_{\text{BB}}(r)$ are plotted in Fig. 1. The first peak of $g_{\text{FeFe}}(r)$ slightly shifts towards right by 0.0014 nm, the difference in $g_{\text{FeSi}}(r)$ and $g_{\text{FeB}}(r)$ is hard to determine. In addition, the difference in the other three PDFs is difficult to be distinguished from the statistic error. So, we think that it cannot cause the marked difference to use a larger cutoff energy than the default one. In the following simulations, we then use such default cutoff energy to save computation time.

When dealing with ferromagnetic elements such as Fe, it is rational to take the electronic spin into consideration. Fig. 1 also includes the PDFs with spin effect at 1473 K. In general, the influence of spin on the PDFs is similar to that of enlarging cutoff energy. Among the three PDFs related with Fe element, only the first peak of $g_{\text{FeFe}}(r)$ is found to shift slightly towards right by 0.003 nm, the difference in $g_{\text{FeSi}}(r)$ and $g_{\text{FeB}}(r)$ is also hard to determine. In Ref. [8], the first peak of the pair distribution function of liquid Fe shifted towards right by 0.007 nm after the electronic spin was taken into consideration, and the $g(r)$ looked more ordered than that without spin effect. In the present work, the influence of spin is suppressed by alloying with Si and B elements. In addition, we think it is difficult to attribute the difference in $g_{\text{SiSi}}(r)$, $g_{\text{SiB}}(r)$ and $g_{\text{BB}}(r)$ to the spin effect. On the scale as in Fig. 1 the PDFs with spin are in good agreement with that of no spin effect. If the spin effect is taken into consideration, the computation time will increase many

times for the same simulation without spin effect. So, in the following simulations, no spin effect is taken into consideration.

A cubic cell containing 77 atoms (60 Fe atoms, 7 Si atoms and 10 B atoms) with periodic boundary condition was used to simulate the liquid system whose composition is actually 77.9221% Fe, 9.0909% Si and 12.9870% B, very close to the composition of commercial $\text{Fe}_{78}\text{Si}_9\text{B}_{13}$ alloy. We employed the same density $91/\text{nm}^3$ during the simulations. Generally only the Γ -point was used to sample the supercell Brillouin zone. All the simulations were carried out in the canonical ensemble (NVT) through a Nosé [9] thermostat with a characteristic frequency equals 52 ps^{-1} . Newton's equations of motion were integrated using the Verlet algorithm with a time step of 3 fs. The system was equilibrated at $T = 1473 \text{ K}$ for 9 ps with 2000 configurations from the last 6 ps period being collected for structural analysis. Then the last configuration was quenched at a cooling rate of $3.33 \times 10^{14} \text{ K/s}$ towards 300 K. At 1373, 1273, 1173, 1073, 973, 873, 773 K, the configurations were simulated for 6 ps to provide structural information.

3. Results and discussion

3.1. Partial pair correlation functions

Since the $g_{\text{FeFe}}(r)$ and $g_{\text{FeSi}}(r)$ are rather normal, i.e. not rich in structural information, we only present the other four PCFs in this

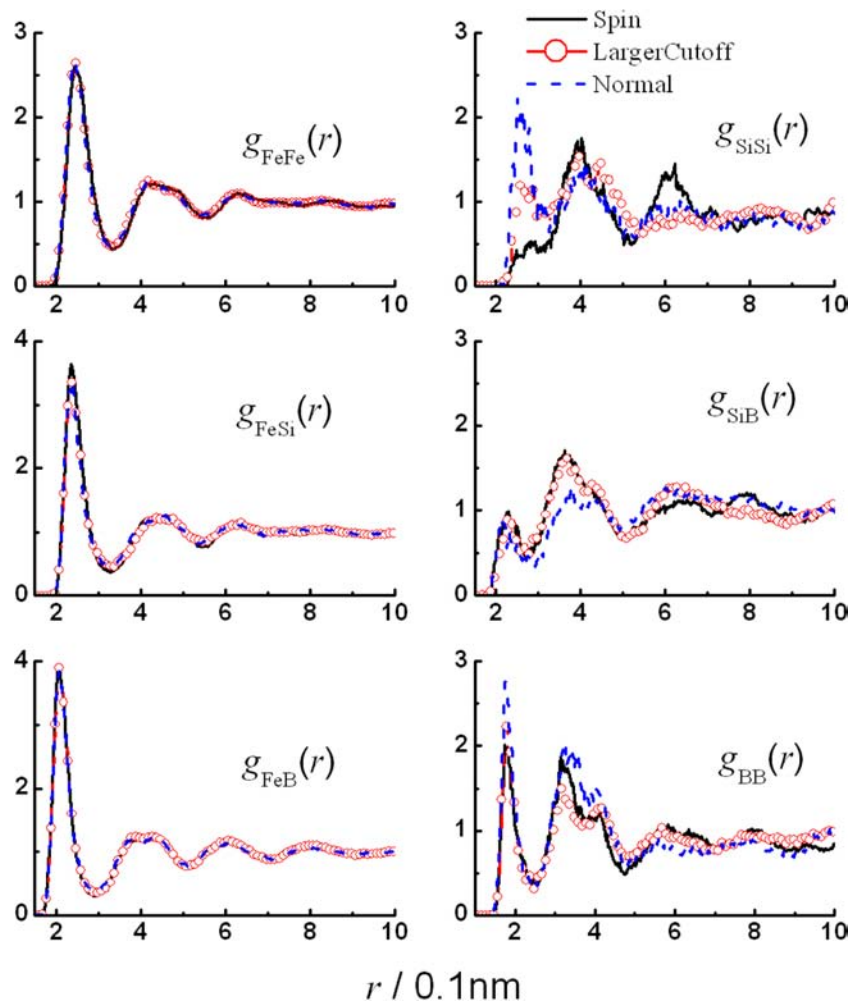


Fig. 1. PDFs of liquid $\text{Fe}_{78}\text{Si}_9\text{B}_{13}$ alloy at 1473 K for three different situations: default cutoff energy (blue dash, labeled as Normal), 1.2 times of the default energy (red line plus open circle, labeled as LargerCutoff), default cutoff energy with spin effect (black line, labeled as Spin). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

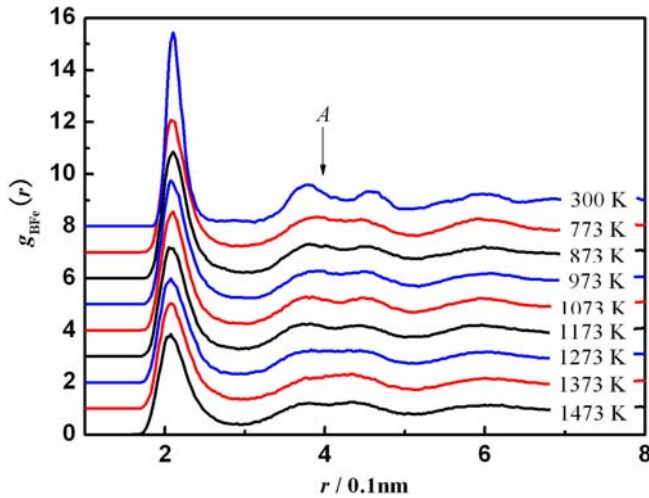


Fig. 2. $g_{BFe}(r)$ of the $Fe_{78}Si_9B_{13}$ alloy at different temperatures.

investigation. Fig. 2 shows the $g_{BFe}(r)$'s of the $Fe_{78}Si_9B_{13}$ alloy at different temperatures. At 300 K, it is found that the calculated PCFs are sensitive to the initio configurations employed. We think it is difficult to get reliable PCFs by quenching only the last configuration of 773–300 K. So we cited the PCFs at 300 K from Ref. [4] in which the PCFs at 300 K were the averaged curves of the six independent quenching routes from 1473 to 300 K.

From Fig. 2 we find that the splitting of the second peak can be perceived at 1473 K and becomes more pronounced with temperature decreasing. At 300 K the second peak around 0.4 nm (indicated by A arrow) has obviously splitted into two subpeaks. The averaged partial coordination number around B for all temperatures is about 8.9. This means that it may be not necessary to associate the splitting of the second peak with the icosahedra clusters.

Fig. 3 shows the PCF of $g_{BSi}(r)$ at different temperatures. This function has a first peak around 0.22 nm (indicated by B arrow), which is relatively minor compared with the second peak (indicated by C arrow) at most temperatures, so it can be neglected except at 873 and 300 K. This peak implies that there are a few B–Si bonds in liquid $Fe_{78}Si_9B_{13}$ alloy. The maximum B–Si coordination number is estimated as 0.37 at 873 K. So, in fact it is rare to have Si–B bonds in both liquid and amorphous alloys. As the temperature decreases, the second peak becomes more pronounced, and the splitting is enhanced.

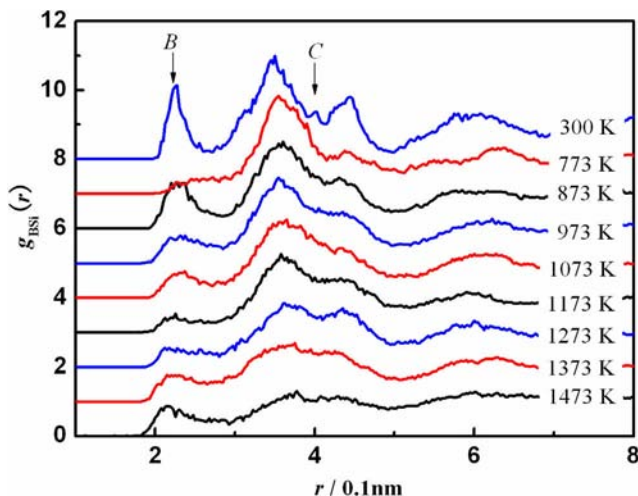


Fig. 3. $g_{BSi}(r)$ of the $Fe_{78}Si_9B_{13}$ alloy at different temperatures.

The PCF of $g_{BB}(r)$ at different temperatures are shown in Fig. 4. Around 0.18 nm (indicated by D arrow), there is weak peak in each curve, which gives the partial coordination number varying from 0.61 for 1473 K to 0.007 for 300 K. Then we can deduce that when the amorphous alloy is formed it is rare for a B atom to meet another B atom. However the most important feature is that the so called second peak (indicated by E arrow) is asymmetric at 1473 K and become more and more splitting with temperature decreasing. At 300 K the second peak evaluates into two well separated peaks with the minimum between them nearly reaching the axis (i.e. at $r_{min} = 0.37$ nm, $g_{BB}(r_{min}) \approx 8$).

Fig. 5 shows the PCF of $g_{SiSi}(r)$ at different temperatures. From 1473 to 873 K both the second peak and the third one (indicated by G and H arrows, respectively) are similar among these curves. However at 1173 K the $g_{SiSi}(r)$ is nearly structureless. At 773 K the second peak (indicated by F arrow) deviates much from that (indicated by G arrow) at the higher temperatures. We think the reason is that the atomic dynamics becomes rather slow and is not ergodic in the configuration space at this temperature. So the correlation between the initio configuration and the following configurations is very strong. This is just the reason why we employed the configuration at 300 K from the previous work [4] and no simulation were performed below 773 K.

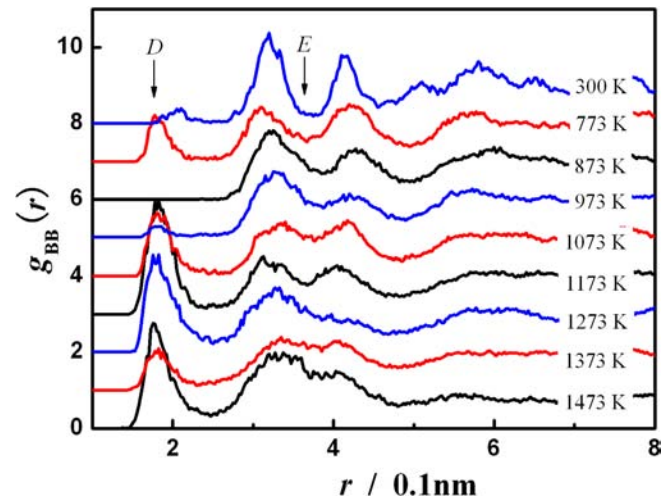


Fig. 4. $g_{BB}(r)$ of the $Fe_{78}Si_9B_{13}$ alloy at different temperatures.

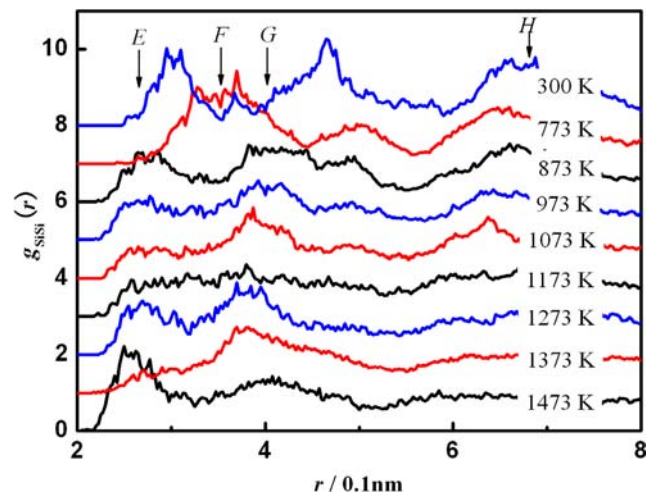


Fig. 5. $g_{SiSi}(r)$ of the $Fe_{78}Si_9B_{13}$ alloy at different temperatures.

In Fig. 5 the first peak at about 0.26 nm (indicated by *E* arrow) may result from Si–Si bonding. The second peak (indicated by *G* arrow) which takes the dominate role in this alloy is located around 0.4 nm at high temperatures. Among the partial coordination number around a Si atom Fe atoms take a proportion up to about 92%. So the second peak is thought resulting from the Si-centered Si–Fe cluster. However, the curve at 773 K is abnormal compared with the rest since its first peak locates at 0.35 nm (indicated by *F* arrow). At 300 K the large deviation is also observed compared with those at higher temperatures.

Since the PCFs associated with B are more regular than those with Si, we investigate the chemical environment around B atoms. The partial coordination numbers around B atoms are listed in Table 1. At all temperatures studied, at least 90% of the atoms around one B atom are Fe atoms, which is much larger than the random distribution (78%). On the other hand the distribution of B and Si atoms around one B atom is much less than their random distribution 13% and 9%, respectively. This chemical environment and the total coordination number around B atoms allow us to make the assumption that statistically one B atom is surrounded by about nine Fe atoms.

So far we have analyzed four kinds of PCFs of the Fe₇₈Si₉B₁₃ alloy, and in Fig. 6 we provide the total correlation functions *g*_{total}(*r*) at different temperatures. We find that the splitting of the second peak is not obvious in *g*_{total}(*r*) above 1273 K. In fact, Mattern et al. [10] did not observe the splitting of the second peak in the total pair correlation function of liquid Fe₈₀B₂₀ and Fe₇₅B₂₅ alloys at 1573 K by X-ray diffraction experiment. All the curves have no peak when *r* is around 0.2 nm except that at 300 K a small shoulder appears. We speculate that if we could have determined the PCFs of the Fe₇₈Si₉B₁₃ alloy by experimental methods, the first peak of *g*_{BB}(*r*) should not have been generated accurately, since both the X-ray scattering factor of B atom and its concentration are small.

3.2. Voronoi analysis

The local structural information can be made clearer by Voronoi analysis [11,12]. Shown in Fig. 7 is the averaged distribution of Voronoi polyhedron around a B atom. In general, for the entire temperatures investigated the peaks are located at 9, which means that a B atom may have about nine coordinating atoms. Taking Table 1 into consideration, we speculate that most B atoms may associate nine Fe atoms to form a cluster with the B atom located at the center. Whether in the liquid state or in the amorphous state this structural feature is dominant in the Fe₇₈Si₉B₁₃ alloy.

In the Fe–B binary phase diagram there is Fe₂B phase [13] in which eight Fe atoms to form an anti-Actinide prism with two B atoms attached near the upper and bottom faces. But during crystallization of Fe₇₈Si₉B₁₃ amorphous alloy, the metastable Fe₃B phase appears at the first stage. In crystalline Fe₃B [14] phase six Fe atoms form an imperfect prism with a B atom located in the center and another three Fe atoms stick to the three four-sided faces.

Table 1
The constitutional proportion (%) of elements in partial coordination number of B atoms.

Temperature/K	Fe (%)	Si (%)	B (%)
1473	90.0	3.3	6.7
1373	93.3	2.7	4.0
1273	90.9	2.6	6.5
1173	91.8	1.5	6.8
1073	94.0	2.4	3.6
973	96.3	3.2	0.5
873	95.9	4.1	0.0
773	96.1	1.8	2.1
300	95.3	4.0	0.7

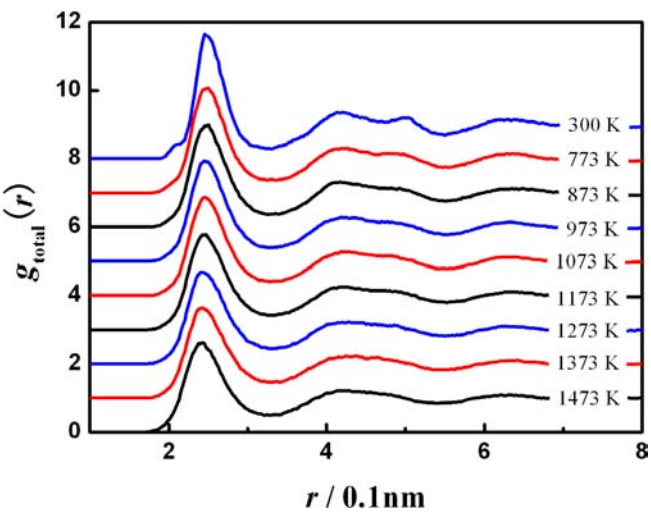


Fig. 6. *g*_{total}(*r*) of the Fe₇₈Si₉B₁₃ alloy at different temperatures.

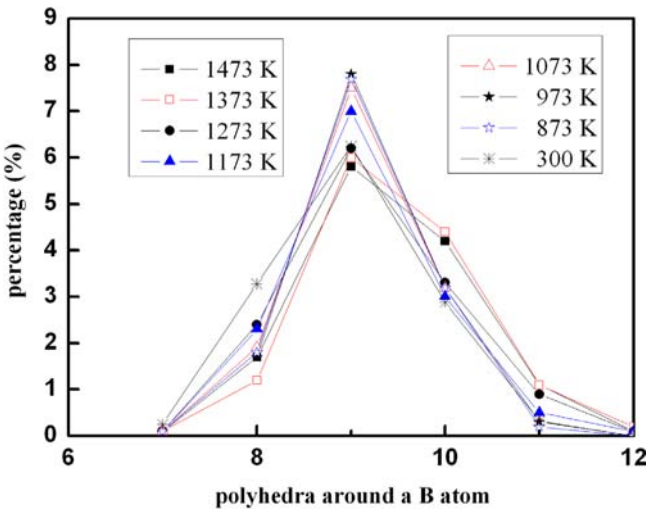


Fig. 7. The distribution of the Voronoi polyhedron around B at different temperatures.

Mattern et al. [10] suggested a model based on such crystalline Fe₃B unit to reproduce the liquid Fe₇₅B₂₅ structure satisfactorily. Based on AIMD simulation we think that the same structural feather around B atoms dominate in both liquid and amorphous Fe-rich binary alloys and Fe₇₈Si₉B₁₃ alloy.

To provide detailed information, we compile the index of Voronoi polyhedron in descending order according to their abundance for the highest temperature studied, then fix the index order as the horizontal axis, and plot the abundance of these indexes at each temperature to see their variation with temperature. In this way, the local structural change will be described not in an averaged manner but rather explicitly. The validity of such method is based on the fact that in these alloys only several tens indexes can weight about 80% of the total amount of the entire indexes. We think this method is more informative than the distribution of Voronoi polyhedron in describing the structural variation with temperature.

In Figs. 8–10 the variation of Voronoi index with temperature is shown for Fe, Si and B atoms respectively. Fig. 8 shows that at temperatures higher than 300 K, the most abundant index is (0, 1, 10, 2), and (0, 2, 8, 2) also has a large proportion. After the system reaches the amorphous state at 300 K, these two important indexes disappear. Some other indexes also disappear which have rather large proportion above 300 K. The index (0, 0, 12, 2) is the

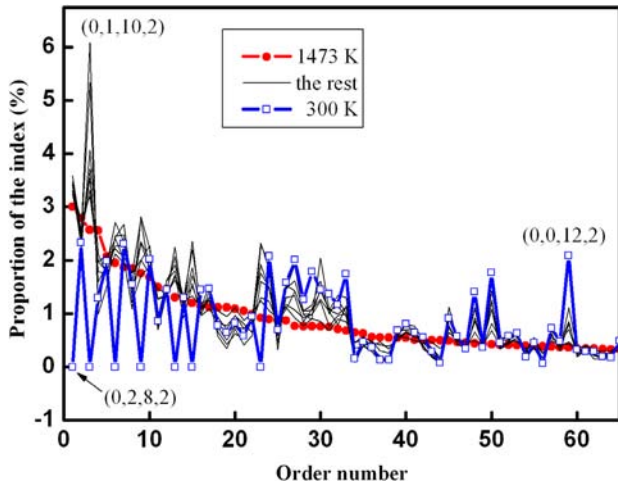


Fig. 8. Comparison among Voronoi indexes of Fe atom at different temperatures.

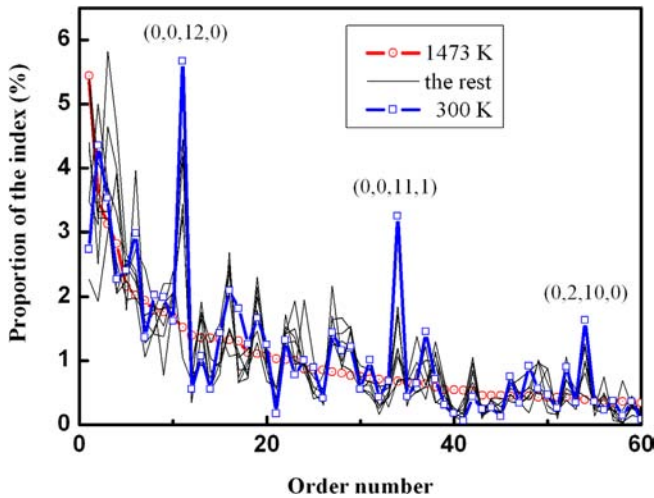


Fig. 9. Comparison among Voronoi indexes of Si atoms at different temperatures.

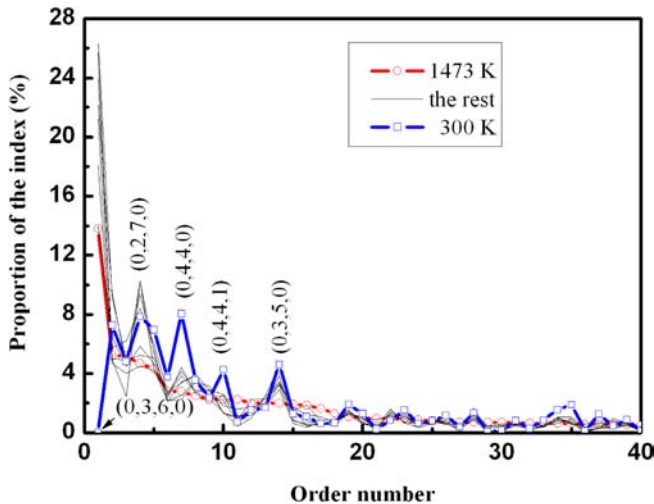


Fig. 10. Comparison among Voronoi indexes of B atoms at different temperatures.

most promoted index which is trivial above 300 K but nearly doubled at 300 K. Around Fe atoms there are few full icosahedra. This may be caused by the penetration of B atoms into the first coordination shell of Fe atoms.

Shown in Fig. 9 is the variation of the abundant indexes of Si atoms at varied temperatures. The most noticeable and consistent change is observed on index (0, 0, 12, 0), (0, 0, 11, 1), (0, 2, 10, 0) below 1473 K. These indexes are full icosahedrons and distorted icosahedrons. Since Si and Fe atoms have similar atomic radii and Si atoms have hardly any B atoms in the first coordination shell, the geometric environment around Si atoms is less distorted than that around Fe atoms. In the amorphous state at 300 K, these three indexes become dominant.

Fig. 10 shows the variation of the abundant indexes of B atoms with temperature. The abundance is concentrated on much less indexes than both Fe and Si atoms. Whether above or at 300 K the indexes (0, 2, 7, 0) and (0, 3, 5, 0) are both important. At 300 K the indexes (0, 4, 4, 0) and (0, 4, 4, 1) are both promoted largely. The most significant change is observed for (0, 3, 6, 0) which becomes zero at 300 K even though it is the most important index at the rest temperatures.

In our simulation the local structure around B atoms in the liquid and amorphous state resembles that of crystalline Fe_3B . In the crystalline Fe_3B , the environment around B could be described by Voronoi index of (0, 3, 6, 0). After transition into amorphous state, this index has disappeared and may be taken place by the series of (0, 2, 7, 0) and (0, 4, 4, 1) which could be viewed as the distorted (0, 3, 6, 0) polyhedron.

3.3. Diffusion coefficients

The diffusion coefficients are valuable to understand the rapid quenching process. We studied the diffusion of atoms in the liquid by calculating the time-dependent mean-square displacement (MSD) defined in the usual way for species α as

$$\langle \Delta r(t)^2 \rangle = \frac{1}{N_\alpha} \left\langle \sum_{i=1}^{N_\alpha} |r_{\alpha i}(t + t_0) - r_{\alpha i}(t_0)|^2 \right\rangle, \quad (1)$$

where the sum goes over all N_α atoms of species α , t_0 is an arbitrary time origin, and the angular brackets denote a thermal average or equivalently an average over time origins. For diffusing liquids the MSD is linear in t for large time, and the slope is proportional to the diffusion coefficient D_α of species α :

$$\langle \Delta r(t)^2 \rangle \rightarrow 6D_\alpha t + B_\alpha, \quad (2)$$

where B_α is constant.

Fig. 11(a) and (b) shows the MSD of Si and B atoms, respectively, of the $\text{Fe}_{78}\text{Si}_9\text{B}_{13}$ alloy before 200 fs. The common feature is that there are minima at about 100 fs in Fig. 11(a) and at about 60 fs in Fig. 11(b). Such phenomenon is thought as the cage-effect with Si and B atoms [15] resulted from the Si and B centered clusters.

The MSD curves after 200 fs were used to fit the diffusion coefficients which are plotted in Fig. 12. We find that Fe and Si atoms have nearly the same diffusion coefficients from 1473 to 300 K. Perhaps this is because their atomic radii are close to each other. Since the B atoms are much smaller than the Fe atoms, they should have much larger diffusion coefficient than that of Fe atoms. In fact, this is just the case above 1173 K. Moreover, during rapid quenching the diffusion coefficient of B atoms decreases more rapidly than those of Fe and Si atoms, and from 1073 K down to 300 K, the three elements have nearly the same diffusion coefficients. We think that from 1073 to 873 K the system is in the supercooled state, since the coherence of the diffusion coefficients in this temperature range implies that the cooperative motion in the alloy may be dominate [15].

From 873 K down to 300 K the diffusion coefficients are only 3–5% of that at 1473 K, one or two orders less than that above 873 K. Therefore, we think the glass transition takes place near 873 K. The diffusion of B atoms at 773 K is abnormal and is not taken into

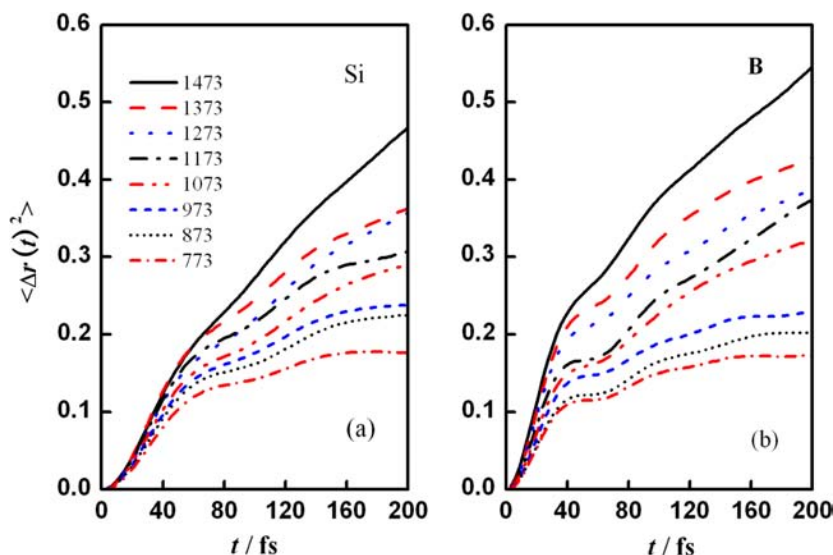


Fig. 11. The initial stage of the MSD of Si and B atoms: (a) MSD for Si atom and (b) MSD for B atom.

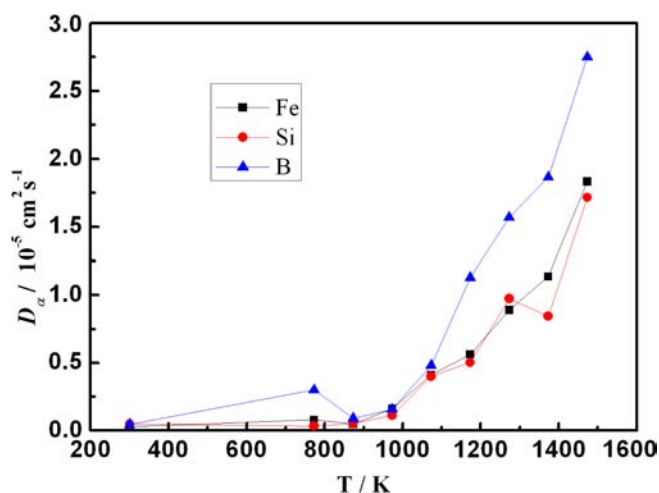


Fig. 12. The diffusion coefficients of Fe, Si and B atoms at different temperatures.

consideration. According to Ref. [16] the crystallization temperature is 813 K which is lower than the glass transition temperature 873 K, so this explains why one cannot observe glass transition by DSC at a heating rate of 20 K/min.

Based on the above discussion, we can understand further the PCF of $g_{\text{SiSi}}(r)$ in Fig. 5. Except at the starting temperature 1473 K, during quenching the first peak at 0.26 nm (indicated by *E* arrow) is depressed at all the lower temperatures. This phenomenon is not observed for both Fe and B atoms. Since Si is minor element in concentration in this alloy, the correlation among the minor elements is perhaps easily influenced by the quenching process. The clustering effect is enhanced to some extent since the second peak near 0.4 nm (indicated by *G* arrow) becomes dominant over the first peak (indicated by *E* arrow). Once the system is vitrified, the atomic motion becomes very slow, and the simulated PCFs of the minor element may depend on the initial configurations. That may be the reason why the PCF of $g_{\text{SiSi}}(r)$ at 773 K deviates largely with those at higher temperatures. We suggest that polymorphous phenomenon exists in respect of correlation among atoms of minor element in quenched $\text{Fe}_{78}\text{Si}_9\text{B}_{13}$ alloy. So in order to model the glassy structure of alloys, the avenue taken in Ref. [4] is more proper than the one employed in this work.

4. Conclusion

Structural information of the $\text{Fe}_{78}\text{Si}_9\text{B}_{13}$ alloy during rapid quenching was derived by AIMD simulation. The second peak splitting has been perceived even at 1473 K in the partial pair correlation functions though not in the total pair correlation function. With temperature decreasing, full icosahedra are rich around Si atoms. The (0, 3, 6, 0) polyhedra are abundant in the liquid state while the distorted (0, 3, 6, 0) polyhedra are the featured local structure around B atoms in the amorphous state.

The diffusion coefficients of Fe and Si are nearly the same at the entire temperature range. B atoms have much larger diffusion coefficient than Fe and Si atoms above 1173 K, but it decreases more rapidly than that of Fe and Si atoms. And from 1173 to 873 K the three coefficients are coherent. We think this temperature range corresponds to the supercooled liquid region, and 873 K serves as the glass transition temperature.

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